| SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE | | DEPARTMENT OF COMPUTER SCIENCE ENGINEERING | |
|---|---|---|---|
| **Program Name:** B. Tech | | **Assignment Type: Lab** | **Academic Year:**2025-2026 |
| **Course Coordinator Name** | Dr. Rishabh Mittal | | |
| **Instructor(s) Name** | Mr. S Naresh Kumar<br>Ms. B. Swathi<br>Dr. Sasanko Shekhar Gantayat<br>Mr. Md Sallauddin<br>Dr. Mathivanan<br>Mr. Y Srikanth<br>Ms. N Shilpa<br>Dr. Rishabh Mittal (Coordinator)<br>Dr. R. Prashant Kumar<br>Mr. Ankushavali MD<br>Mr. B Viswanath<br>Ms. Sujitha Reddy<br>Ms. A. Anitha<br>Ms. M.Madhuri<br>Ms. Katherashala Swetha<br>Ms. Velpula sumalatha<br>Mr. Bingi Raju | | |
| **CourseCode** | 23CS002PC304 | **Course Title** | AI Assisted Coding |
| **Year/Sem** | III/II | **Regulation** | R23 |
| **Date and Day of Assignment** | **Week10 – Wednesday** | **Time(s)** | 23CSBTB01 To 23CSBTB52 |
| **Duration** | 2 Hours | **Applicable to Batches** | All batches |
| **Assignment Number:19.3**(Present assignment number)/**24**(Total number of assignments) | | | |

| Q.No. | Question | *Expected Time to complete* |
|---|---|---|
| 1 | **Lab 19 – Code Translation: Converting Between Programming Languages**<br>**Lab Objectives:**<br>• Understand how AI tools can assist in translating code between | Week10 - Wednesday |

different programming languages.

- Learn to verify correctness and functionality after translation.
- Explore syntactic and semantic differences between languages (e.g., Python, Java, C++).
- Practice debugging and optimizing AI-translated code.

**Task 1 – Python to C++ Conversion**

**Task:**

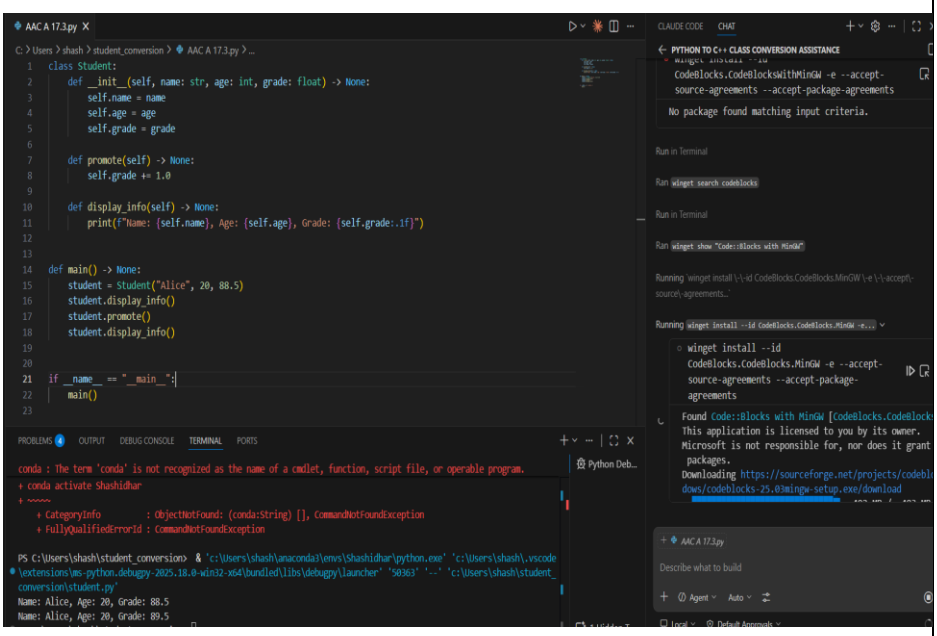
Translate a simple Python class into C++ using AI assistance.

**Instructions:**

- Prompt AI to convert a Python class representing a Student with attributes and methods into equivalent C++ code
- Ensure proper handling of:
  - Constructors
  - Data types
  - Access specifiers
- Compile and run both versions to verify output consistency

**Expected Output:**

An equivalent working C++ class translated from Python.




**Task 2 – Java to Python Function Conversion**

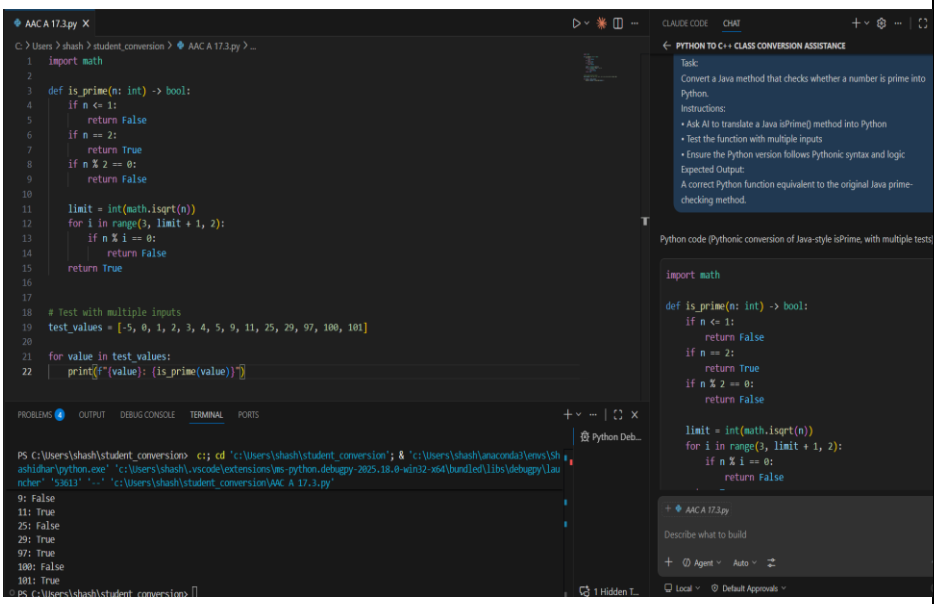
**Task:**

Convert a Java method that checks whether a number is prime into Python.

**Instructions:**

- Ask AI to translate a Java isPrime() method into Python
- Test the function with multiple inputs
- Ensure the Python version follows Pythonic syntax and logic

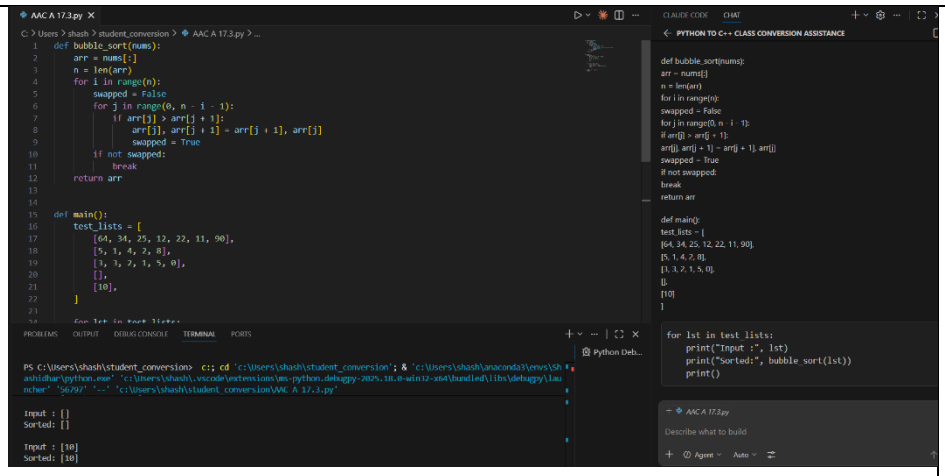
**Expected Output:**
A correct Python function equivalent to the original Java prime-checking method.

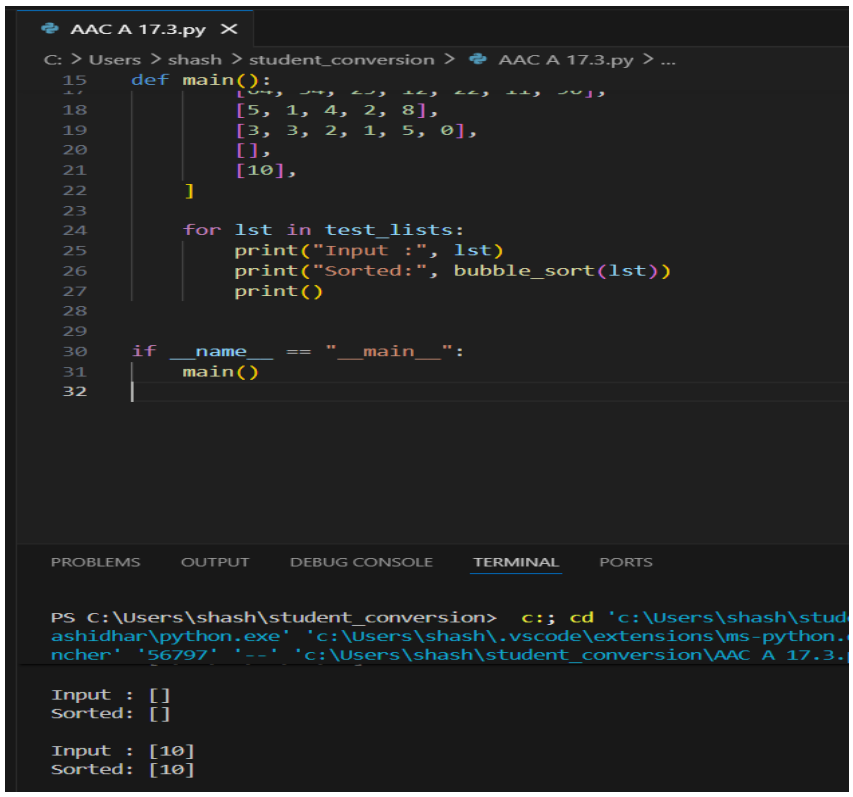



**Task 3 – Pseudocode to Python Implementation**
**Task:**
Translate a given pseudocode algorithm into Python using AI.
**Instructions:**

- Provide AI with pseudocode for sorting numbers (e.g., bubble sort or selection sort)
- Ask AI to convert it into executable Python code
- Validate correctness using sample input lists

**Expected Output:**
A working Python program that correctly implements the pseudocode logic.




**Task 4 – SQL to Pandas Query**

**Task:**

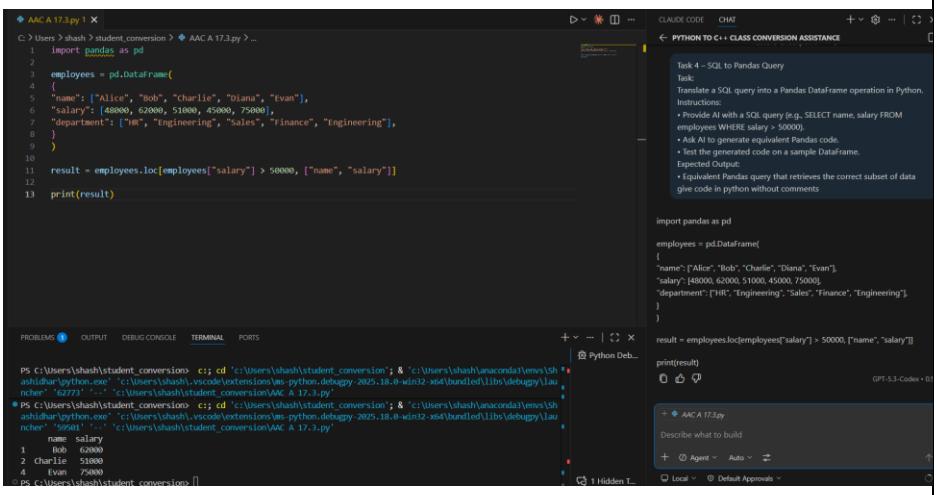
Translate a **SQL query** into a **Pandas DataFrame operation** in Python.

**Instructions:**

- Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

**Expected Output:**

- Equivalent Pandas query that retrieves the correct subset of data.




**Task 5 – Real-Time Application: Algorithm Translation Across Languages**

**Scenario:**

A company maintains algorithms written in different programming languages.

**Examples:**

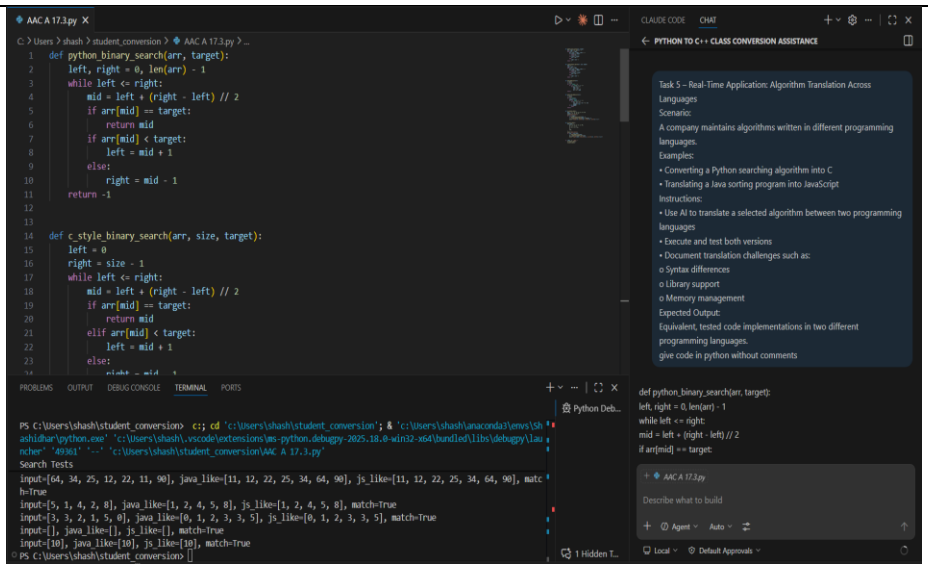
- Converting a Python searching algorithm into C
- Translating a Java sorting program into JavaScript

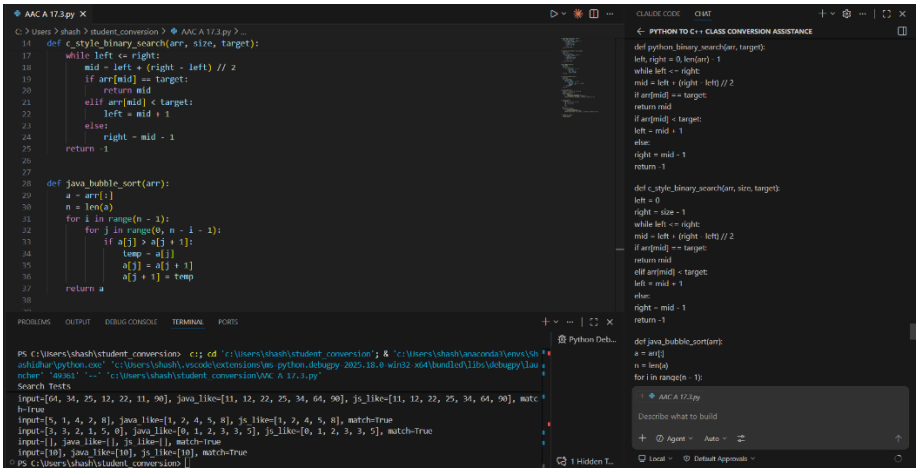
**Instructions:**

- Use AI to translate a selected algorithm between two programming languages
- Execute and test both versions
- Document translation challenges such as:
    - Syntax differences
    - Library support
    - Memory management

**Expected Output:**

Equivalent, tested code implementations in two different programming languages.


AAC A 17.3.py
C: > Users > shash > student_conversion > AAC A 17.3.py > ...
def python_binary_search(arr, target):
left, right = 0, len(arr) - 1
while left <= right:
mid = left + (right - left) // 2
if arr[mid] == target:
return mid
if arr[mid] < target:
left = mid + 1
else:
right = mid - 1
return -1
def c_style_binary_search(arr, size, target):
left = 0
right = size - 1
while left <= right:
mid = left + (right - left) // 2
if arr[mid] == target:
return mid
elif arr[mid] < target:
left = mid + 1
else:
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\shash\student_conversion> c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '49361' '--' 'c:\Users\shash\student_conversion\AAC A 17.3.py'
Search Tests
input=[64, 34, 25, 12, 22, 11, 90], java_like=[11, 12, 22, 25, 34, 64, 90], js_like=[11, 12, 22, 25, 34, 64, 90], match=True
input=[5, 1, 4, 2, 8], java_like=[1, 2, 4, 5, 8], js_like=[1, 2, 4, 5, 8], match=True
input=[3, 3, 2, 1, 5, 0], java_like=[0, 1, 2, 3, 3, 5], js_like=[0, 1, 2, 3, 3, 5], match=True
input=[], java_like=[], js_like=[], match=True
input=[10], java_like=[10], js_like=[10], match=True
PS C:\Users\shash\student_conversion>
Python Deb...
1 Hidden T...
CLAUDE CODE CHAT
PYTHON TO C++ CLASS CONVERSION ASSISTANCE
Task 5 – Real-Time Application: Algorithm Translation Across Languages
Scenario:
A company maintains algorithms written in different programming languages.
Examples:
• Converting a Python searching algorithm into C
• Translating a Java sorting program into JavaScript
Instructions:
• Use AI to translate a selected algorithm between two programming languages
• Execute and test both versions
• Document translation challenges such as:
o Syntax differences
o Library support
o Memory management
Expected Output:
Equivalent, tested code implementations in two different programming languages.
give code in python without comments
def python_binary_search(arr, target):
left, right = 0, len(arr) - 1
while left <= right:
mid = left + (right - left) // 2
if arr[mid] == target:
AAC A 17.3.py
Describe what to build
Agent
Auto
Local
Default Approvals


AAC A 17.3.py
C: > Users > shash > student_conversion > AAC A 17.3.py > ...
def c_style_binary_search(arr, size, target):
while left <= right:
mid = left + (right - left) // 2
if arr[mid] == target:
return mid
elif arr[mid] < target:
left = mid + 1
else:
right = mid - 1
return -1
def java_bubble_sort(arr):
a = arr[:]
n = len(a)
for i in range(n - 1):
for j in range(0, n - i - 1):
if a[j] > a[j + 1]:
temp = a[j]
a[j] = a[j + 1]
a[j + 1] = temp
return a
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\shash\student_conversion> c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '49363' '--' 'c:\Users\shash\student_conversion\AAC A 17.3.py'
Search Tests
input=[64, 34, 25, 12, 22, 11, 90], java_like=[11, 12, 22, 25, 34, 64, 90], js_like=[11, 12, 22, 25, 34, 64, 90], match=True
input=[5, 1, 4, 2, 8], java_like=[1, 2, 4, 5, 8], js_like=[1, 2, 4, 5, 8], match=True
input=[3, 3, 2, 1, 5, 0], java_like=[0, 1, 2, 3, 3, 5], js_like=[0, 1, 2, 3, 3, 5], match=True
input=[], java_like=[], js_like=[], match=True
input=[10], java_like=[10], js_like=[10], match=True
PS C:\Users\shash\student_conversion>
Python Deb...
1 Hidden T...
CLAUDE CODE CHAT
PYTHON TO C++ CLASS CONVERSION ASSISTANCE
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while left <= right:
mid = left + (right - left) // 2
if arr[mid] == target:
return mid
if arr[mid] < target:
left = mid + 1
else:
right = mid - 1
return -1
def c_style_binary_search(arr, size, target):
left = 0
right = size - 1
while left <= right:
mid = left + (right - left) // 2
if arr[mid] == target:
return mid
elif arr[mid] < target:
left = mid + 1
else:
right = mid - 1
return -1
def java_bubble_sort(arr):
a = arr[:]
n = len(a)
for i in range(n - 1):
AAC A 17.3.py
Describe what to build
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Auto
Local
Default Approvals

AAC A 17.3.py

C: > Users > shash > student_conversion > AAC A 17.3.py > ...

```python
def javascript_bubble_sort(arr):
    a = arr[:]
    n = len(a)
    for i in range(n):
        swapped = False
        for j in range(0, n - i - 1):
            if a[j] > a[j + 1]:
                a[j], a[j + 1] = a[j + 1], a[j]
                swapped = True
        if not swapped:
            break
    return a


def run_search_tests():
    data = [3, 7, 12, 19, 24, 31, 45, 58]
    targets = [3, 24, 58, 9]
    print("Search Tests")
    for t in targets:
        p = python_binary_search(data, t)
        c = c_style_binary_search(data, len(data), t)
        print(f"target={t}, python={p}, c_style={c}, match={p == c}")

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\Sh
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\lau
ncher' '49361' '--' 'c:\Users\shash\student_conversion\AAC A 17.3.py'
Search Tests
input=[64, 34, 25, 12, 22, 11, 90], java_like=[11, 12, 22, 25, 34, 64, 90], js_like=[11, 12, 22, 25, 34, 64, 90], matc
h=True
input=[5, 1, 4, 2, 8], java_like=[1, 2, 4, 5, 8], js_like=[1, 2, 4, 5, 8], match=True
input=[3, 3, 2, 1, 5, 0], java_like=[0, 1, 2, 3, 3, 5], js_like=[0, 1, 2, 3, 3, 5], match=True
input=[], java_like=[], js_like=[], match=True
input=[10], java_like=[10], js_like=[10], match=True
```

AAC A 17.3.py

C: > Users > shash > student_conversion > AAC A 17.3.py > ...

```python

def run_sort_tests():
    test_cases = [
        [64, 34, 25, 12, 22, 11, 90],
        [5, 1, 4, 2, 8],
        [3, 3, 2, 1, 5, 0],
        [],
        [10],
    ]
    print("\nSort Tests")
    for case in test_cases:
        j = java_bubble_sort(case)
        js = javascript_bubble_sort(case)
        print(f"input={case}, java_like={j}, js_like={js}, match={j == js}")


if __name__ == "__main__":
    run_search_tests()
    run_sort_tests()

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs
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h=True
input=[5, 1, 4, 2, 8], java_like=[1, 2, 4, 5, 8], js_like=[1, 2, 4, 5, 8], match=True
input=[3, 3, 2, 1, 5, 0], java_like=[0, 1, 2, 3, 3, 5], js_like=[0, 1, 2, 3, 3, 5], match=True
input=[], java_like=[], js_like=[], match=True
input=[10], java_like=[10], js_like=[10], match=True
```

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.